

Solutions:

1. B. 3

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 2x + 1}{x^2 + x + 1} = \frac{3(\infty^2) + 2(\infty) + 1}{\infty^2 + \infty + 1} = \frac{\infty}{\infty} \text{ (indeterminate form)}$$

Apply L'Hospital's Rule

$$\frac{\frac{d}{dx}(3x^2 + 2x + 1)}{\frac{d}{dx}(x^2 + x + 1)} = \frac{6x + 2}{2x + 1}$$

Let $x = \infty$

$$\frac{6(\infty) + 2}{2(\infty) + 1} = \frac{\infty}{\infty} \text{ (indeterminate form)}$$

Apply L'Hospital's Rule:

$$\frac{\frac{d}{dx}(6x + 2)}{\frac{d}{dx}(2x + 1)} = \frac{6}{2} = 3$$

Calculator Shortcut: Using Casio FX570ES+

Assign a value of 'x' very close to ∞ .

$$\frac{3x^2 + 2x + 1}{x^2 + x + 1} \rightarrow \text{CALC} \rightarrow x = 9999999 \rightarrow \boxed{=3}$$

2. A. $y = -x + 2$

$$\text{At } x = 1, y = 1$$

$$y = x^3 - 2x^2 + 2$$

$$\frac{dy}{dx} = \frac{d}{dx}(x^3 - 2x^2 + 2) \Big|_{x=1}$$

Calculating:

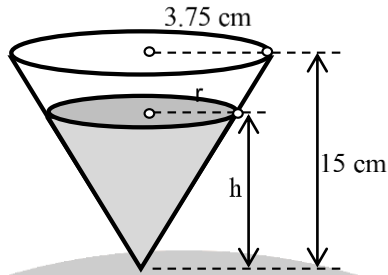
$$\frac{dy}{dx} = -1$$

$$\text{So, } y - y_1 = m(x - x_1)$$

$$y - 1 = -1(x - 1)$$

$$y = -x + 2$$

3. B. $6.28 \text{ m}^3/\text{min}$



$$V = \frac{\pi r^2 h}{3}$$

$$\frac{r}{h} = \frac{3.75}{15}$$

$$r = \frac{h}{4}$$

$$V = \frac{\pi r^2 h}{3} = \frac{\pi (h^2) h}{3(16)} = \frac{\pi h^3}{48}$$

$$\frac{dV}{dt} = \frac{\pi}{48} (3) h^2 \frac{dh}{dt}$$

$$\frac{dV}{dt} = \frac{\pi}{16} (4)^2 (2)$$

$$\frac{dV}{dt} = 6.28 \text{ m}^3/\text{min.}$$

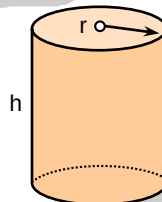
4. A. 125.50

$$V = \pi r^2 h$$

$$108 = \pi r^2 h$$

$$h = \frac{108}{\pi r^2} \rightarrow \text{Eq.1}$$

$$A = 2\pi r h + 2(\pi r^2) \rightarrow \text{Eq.2}$$



Substitute Eq.1 in Eq.

$$A = 2\pi r \left(\frac{108}{\pi r^2} \right) + 2(\pi r^2)$$

$$A = \frac{216}{r} + 2\pi r^2 \rightarrow \text{Eq.3}$$

Differentiate

$$\frac{dA}{dr} = \frac{-216}{r^2} + 4\pi r$$

$$0 = \frac{-216}{r^2} + 4\pi r$$

$$\frac{216}{r^2} = 4\pi r$$

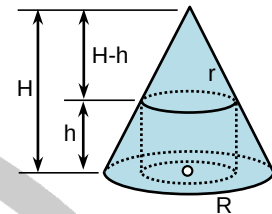
$$r^3 = \frac{216}{4\pi}$$

$$r = 2.58 \text{ in}$$

Substitute r in Eq.3:

$$A = \frac{216}{2.58} + 2\pi(2.58)^2 = 125.544 \text{ in}^2$$

5. A. 44 %



Let:

R = radius of the cone

H = height of the cone

r = radius of the inscribed cylinder

h = height of the inscribed cylinder

V = volume of cylinder

$$V = \pi r^2 h \rightarrow \text{Eq.1}$$

By ratio and proportion:

$$\frac{R}{H} = \frac{r}{H-h}$$

$$Hr = RH - Rh$$

$$h = \frac{RH - Hr}{R} \rightarrow \text{Eq.2}$$

Substitute Eq.2 in Eq.1:

$$V = \pi r^2 \left(\frac{RH - Hr}{R} \right) = \pi r^2 H - \frac{\pi r^3 H}{R}$$

Differentiate:

$$\frac{dV}{dr} = 2\pi r H - \frac{3\pi r^2 H}{R}$$

$$0 = 2\pi r H - \frac{3\pi r^2 H}{R}$$

$$\frac{3\pi r^2 H}{R} = 2\pi r H$$

$$r = \frac{2}{3} R$$

Substitute r in Eq.2:

$$h = \frac{RH - H(2/3 R)}{R} = \frac{1}{3} H$$

Substitute r and h in Eq.1:

$$V = \pi \left(\frac{2}{3} R \right)^2 \left(\frac{1}{3} H \right) = \pi \left(\frac{4}{27} \right) R^2 H$$

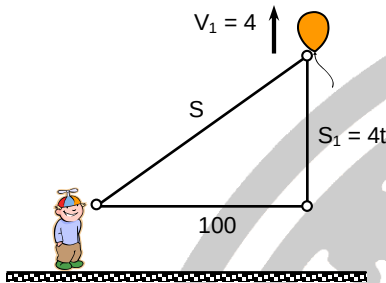
Let: V_C = volume of the cone



$$V_c = \frac{1}{3} \pi R^2 H$$

$$\text{Ratio} = \frac{V}{V_c} = \frac{\pi \left(\frac{4}{27} \right) R^2 H}{\frac{1}{3} \pi R^2 H} = 0.44$$

6. D. 1.49 m/sec



$$S^2 = 100^2 + (S_1)^2$$

$$S^2 = 100^2 + (4t)^2$$

$$S^2 = 10,000 + 16t^2 \rightarrow \text{Eq. 1}$$

Differentiate both sides with respect to t:

$$2S \left(\frac{dS}{dt} \right) = 0 + 32t \left(\frac{dt}{dt} \right)$$

$$\frac{dS}{dt} = \frac{16t}{S} \rightarrow \text{Eq. 2}$$

Substitute t = 10 in Eq. 1:

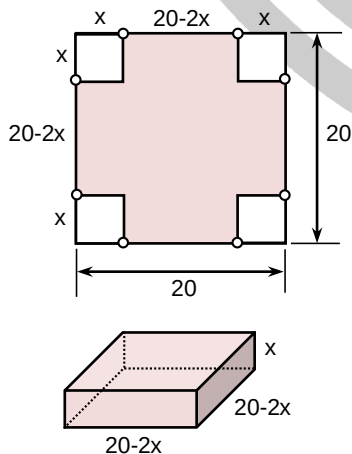
$$S^2 = 10,000 + 16(10)^2$$

$$S = 107.7 \text{ m}$$

Substitute t = 10 and S = 107.7 in Eq. 2:

$$\frac{dS}{dt} = \frac{16(10)}{107.7} = 1.485 \text{ m/s}$$

7. B. 592.59 cu in.



$$V = (20 - 2x)(20 - 2x)(x)$$

$$V = (20 - 2x)^2 (x)$$

$$V = (400 - 80x + 4x^2)(x)$$

$$V = 400x - 80x^2 + 4x^3 \rightarrow \text{Eq. 1}$$

Differentiating:

$$\frac{dV}{dx} = 400 - 160x + 12x^2$$

$$0 = 400 - 160x + 12x^2$$

By factoring:

$$(4x - 40)(3x - 10) = 0$$

$$4x = 40$$

$$x = 10 \text{ in (absurd)}$$

$$3x = 10$$

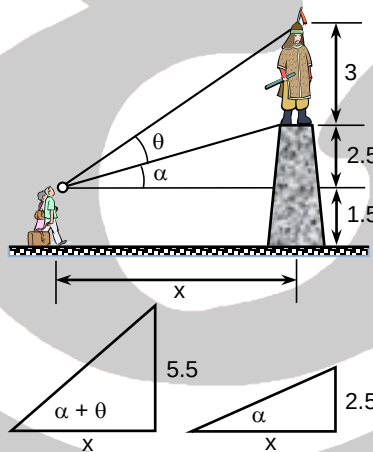
$$x = 3.33 \text{ in}$$

Substituting x = 3.33 in Eq. 1

$$V = 400(3.33) - 80(3.33)^2 + 4(3.33)^3$$

$$V = 592.59 \text{ in}^3$$

8. C. 3.71 m



$$\tan(\alpha + \theta) = \frac{5.5}{x}$$

$$\frac{\tan \alpha + \tan \theta}{1 - \tan \alpha \tan \theta} = \frac{5.5}{x}$$

$$\frac{\frac{2.5}{x} + \tan \theta}{1 - \frac{2.5}{x} \tan \theta} = \frac{5.5}{x}$$

$$2.5 + x \tan \theta = 5.5 - \frac{13.75}{x} \tan \theta$$

$$\tan \theta \left(x + \frac{13.75}{x} \right) = 3$$

$$\tan \theta = \frac{3}{x + \frac{13.75}{x}}$$

$$\tan \theta = \frac{3x}{x^2 + 13.75}$$

$$\theta = \tan^{-1} \left(\frac{3x}{x^2 + 13.75} \right)$$

$$\text{Note: } d(\tan^{-1} u) = \frac{du}{1 + u^2}$$

where:

$$u = \frac{3x}{x^2 + 13.75}$$

$$u^2 = \frac{9x^2}{(x^2 + 13.75)^2}$$

$$du = \frac{(x^2 + 13.75)3 - 3x(2x)}{(x^2 + 13.75)^2}$$

Substituting:

$$\frac{d\theta}{dx} = \frac{du}{1 + u^2}$$

$$0 = \frac{du}{1 + u^2}$$

$$0 = du$$

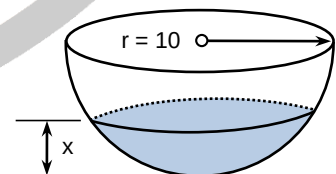
$$0 = \frac{(x^2 + 13.75)3 - 3x(2x)}{(x^2 + 13.75)^2}$$

$$2x^2 = x^2 + 13.75$$

$$x^2 = 13.75$$

$$x = 3.708$$

9. C. 4



$$V = \frac{\pi x^2}{3} (3r - x)$$

$$V = \frac{\pi x^2}{3} (3(10) - x)$$

$$V = 10\pi x^2 - \frac{\pi x^3}{3}$$



Differentiating both sides with respect to t:

$$\frac{dV}{dt} = \left(20\pi x - \frac{3\pi x^2}{3} \right) \frac{dx}{dt}$$

$$\frac{dV}{dt} = (20\pi x - \pi x^2) \frac{dx}{dt} \rightarrow \text{Eq.1}$$

Substituting $dV/dt = 3$ and $dx/dt = 0.0149$ in Eq.1:

$$3 = (20\pi x - \pi x^2)(0.0149)$$

$$201.342 = 20\pi x - \pi x^2$$

$$64 = 20x - x^2$$

$$x^2 - 20x + 64 = 0$$

$$(x-4)(x-16) = 0$$

$$x = 4 \text{ cm or}$$

$$x = 16 \text{ cm}$$

10. B. 2/5

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 + 3x - 4}$$

Substitute $x = 1$:

$$\text{Lim} = \frac{(1)^2 - 1}{(1)^1 + 3(1) - 4} = \frac{0}{0}, \text{ indeterminate}$$

Using L'Hopital's rule, differentiate separately the numerator and denominator and substitute the value of limit to the variable

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 + 3x - 4} = \frac{2x}{2x + 3}$$

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 + 3x - 4} = \frac{2(1)}{2(1) + 3}$$

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 + 3x - 4} = \frac{2}{5}$$

11. B. 1/2

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$\text{Lim} = \frac{1 - \cos 0}{(0)^2} = \frac{0}{0}, \text{ indeterminate}$$

Apply L'Hopital's rule & substitute $x = 0$:

$$\text{Lim} = \frac{\sin x}{2x}$$

$$\text{Lim} = \frac{\sin 2(0)}{2(0)} = \frac{0}{0}, \text{ indeterminate}$$

Apply L'Hopital's rule again:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{\cos x}{2}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{\cos 0}{2}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

12. D. $6x + 3$

$$y = (x+1)^3 - x^3$$

Note: $d(u)^n = nu^{n-1} du$

$$y' = 3(x+1)^2(1) - 3(x)^2(1)$$

$$y' = 3(x+1)^2 - 3(x)^2$$

$$y' = 3(x^2 + 2x + 1) - 3x^2$$

$$y' = \cancel{3x^2} + 6x + 3 - \cancel{3x^2}$$

$$y' = 6x + 3$$

13. A. -1

$$y = \sin^{-1} \cos x$$

$$\text{Note: } d(\sin^{-1} u) = \frac{du}{\sqrt{1-u^2}}$$

where:

$$u = \cos x$$

$$u^2 = \cos^2 x$$

$$du = -\sin x$$

$$y' = \frac{-\sin x}{\sqrt{1-\cos^2 x}}$$

$$y' = \frac{-\sin x}{\sqrt{\sin^2 x}}$$

$$y' = \frac{-\sin x}{\sin x}$$

$$y' = -1$$

14. A. $\frac{x^2 + 2x}{(x+1)^2}$

$$y = \frac{x^2}{x+1}$$

$$\text{Note: } d\left(\frac{u}{v}\right) = \frac{vdu - u dv}{v^2}$$

where:

$$u = x^2$$

$$du = 2x$$

$$v = x + 1$$

$$v^2 = (x + 1)^2$$

$$dv = 1$$

$$y' = \frac{(x+1)(2x) - x^2(1)}{(x+1)^2}$$

$$y' = \frac{2x^2 + 2x - x^2}{(x+1)^2}$$

$$y' = \frac{2x^2 + 2x - x^2}{(x+1)^2}$$

$$y' = \frac{x^2 + 2x}{(x+1)^2}$$

15. -1 1/4

$$y = \frac{x^3}{4} - 2x + 1$$

Note: slope = y'

Differentiating and substituting $x = 1$:

$$y' = \frac{1}{4}(3x^2) - 2$$

$$y' = \frac{1}{4}(3)(1)^2 - 2$$

$$y' = -1 \frac{1}{4}$$

16. B. $x = 2y$

$$x^2 + y^2 = 1$$

Let: m_1 = slope of the given curve

Differentiating and substituting $x = 2$ & $y = 1$:

$$2x + 2yy' = 0$$

$$y' = -\frac{x}{y}$$

$$y' = -\frac{2}{1}$$

$$y' = -2$$

$$m_1 = -2$$



Since the line is normal to the curve at the given point, the slope (m_2) of the line is equal the negative reciprocal of the slope of the given curve.

$$m_2 = -\frac{1}{m_1}$$

$$m_2 = -\frac{1}{-2}$$

$$m_2 = \frac{1}{2}$$

Using point slope form:

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{2}(x - 2)$$

$$2y - 2 = x - 2$$

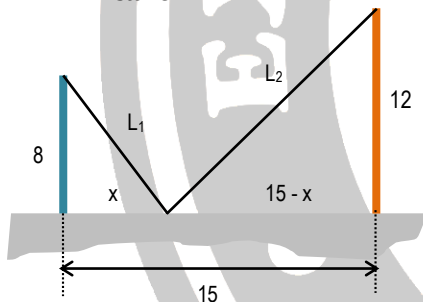
$$x = 2y$$

17. B. $y'' = 0$

18. B. second

19. C. 4 m from the lower post

Let: x = distance from lower post to stake



Total length of cable used, L is:

$$L = L_1 + L_2 \rightarrow \text{Eq. 1}$$

Solving for L_1 using Pythagorean theorem:

$$L_1 = \sqrt{8^2 + x^2}$$

Solving for L_2 using Pythagorean theorem:

$$L_2 = \sqrt{12^2 + (15 - x)^2}$$

Substituting the values of L_1 & L_2 in Eq. 1:

$$L = \sqrt{8^2 + x^2} + \sqrt{12^2 + (15 - x)^2}$$

$$L = \sqrt{64 + x^2} + \sqrt{144 + (15 - x)^2}$$

Differentiating using the following formula:

$$\frac{d}{dx} \sqrt{u} = \frac{du}{2\sqrt{u}}$$

$$\frac{dL}{dx} = \frac{2x}{2\sqrt{64 + x^2}} + \frac{2(15 - x)(-1)}{2\sqrt{144 + (15 - x)^2}}$$

Equating to zero to obtain minimum condition:

$$\frac{2x}{2\sqrt{64 + x^2}} + \frac{2(15 - x)(-1)}{2\sqrt{144 + (15 - x)^2}} = 0$$

$$\frac{15 - x}{\sqrt{144 + (15 - x)^2}} = \frac{x}{\sqrt{64 + x^2}}$$

$$\frac{15 - x}{x} = \frac{\sqrt{144 + (15 - x)^2}}{\sqrt{64 + x^2}}$$

Squaring both sides:

$$\left(\frac{15 - x}{x}\right)^2 = \frac{144 + (15 - x)^2}{64 + x^2}$$

$$(15 - x)^2 (64 + x^2) = x^2 (144 + (15 - x)^2)$$

$$\left[\begin{aligned} (15 - x)^2 (64) \\ + (15 - x)^2 x^2 \end{aligned} \right] = 144x^2 + (15 - x)^2 x^2$$

$$(15 - x)^2 (64) = 144x^2$$

$$(15 - x)^2 = \frac{144x^2}{64}$$

Extracting the square root on both sides:

$$15 - x = \frac{12x}{8}$$

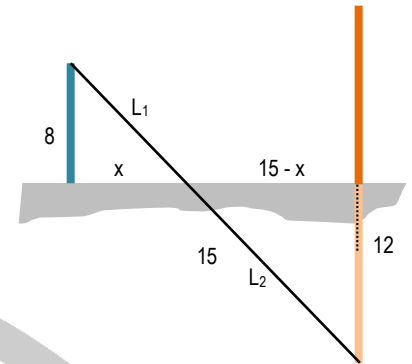
$$120 - 8x = 12x$$

$$120 = 20x$$

$$x = 6$$

Alternate solution:

Create a mirror image of the longer post. The cable connecting the top of shorter post to the stake then to the top of longer post must be a straight line (minimum distance between two points).



Using two similar triangles, by ratio and proportion:

$$\frac{x}{8} = \frac{15 - x}{12}$$

$$12x = 120 - 8x$$

$$20x = 120$$

$$x = 6 \text{ m}$$

20. B. 72

$$x = 18t + 9t^2$$

Since velocity is the first derivative of the distance, differentiating the above equation with respect to time:

$$\frac{dx}{dt} = 18 + 18t$$

$$V = 18 + 18t$$

When $t = 3$:

$$V = 18 + 18(3)$$

$$V = 72$$

21. B. $1/\pi$

$$A = \pi r^2$$

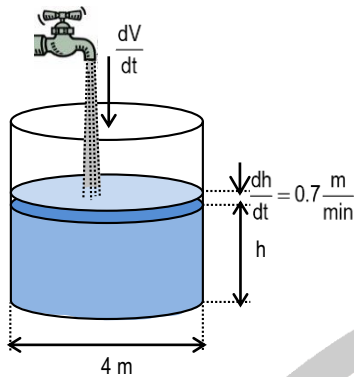
$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} \rightarrow \text{Eq. 1}$$

From given $\frac{dA}{dt} = 2 \frac{dr}{dt}$. Substituting in

Eq 1

$$2 \frac{dr}{dt} = 2\pi r \frac{dr}{dt} \Rightarrow r = \frac{1}{\pi}$$

22. D. 8.7964 cu. m/min



Volume of water in cylinder, V is:

$$V = \pi r^2 h$$

$$V = \pi (2)^2 h$$

$$V = 4\pi h$$

Differentiating with respect to time:

$$\frac{dV}{dt} = 4\pi \frac{dh}{dt}$$

Substituting $\frac{dh}{dt} = 0.7$

$$\frac{dV}{dt} = 4\pi (0.7)$$

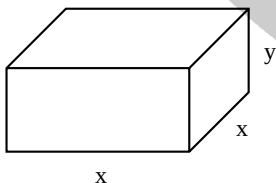
$$\frac{dV}{dt} = 8.7964$$

23. D. 2

$$\ln e^{2x} = 2x$$

$$\frac{d}{dx} (\ln e^{2x}) = \frac{d}{dx} (2x) = 2$$

24. C. 32



Total surface area:

$$x^2 + 4xy = 48$$

$$y = \frac{48 - x^2}{4x} = \frac{12}{x} - \frac{1}{4}x$$

$$V = x^2 y = x^2 \left[\frac{12}{x} - \left(\frac{1}{4} \right) x \right] = 12x - \frac{1}{4}x^3$$

$$V' = 12 - \left(\frac{1}{4} \right) 3x^2 = 0$$

$$x = 4$$

$$y = \frac{12}{4} - \frac{1}{4}(4) = 2$$

$$V = (4)(4)(2) = 32$$

25. A. 4

$$V = \pi r^2 h$$

$$16 = \pi r^2 h$$

$$h = 16r^{-2}$$

$$A = 2\pi r h + 2\pi r^2 = 2\pi (16r^{-1} + r^2)$$

$$\frac{dA}{dr} = 2\pi (-16r^{-2} + 2r) = 4\pi r^{-2} (-8 + r^3)$$

$$\frac{dA}{dr} < 0 \text{ for } 0 < r < 2 \text{ and } \frac{dA}{dr} > 0 \text{ for } r > 2$$

The minimum surface area of the can is when $r = 2$ and $h = 4$